

Agenda

1. Announcements

- 11.8, 11.9, 11.10 HW due Wednesday
 - Discussion Quiz 9 on Thursday over
- 

2. Section 11.11

3.) Section 12.1

Approximating Functions

Suppose $f(x)$ has the Taylor series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n = f(c) + f'(c) + \frac{f''(c)}{2} (x-c)^2 + \frac{f'''(c)}{6} (x-c)^3 + \dots$$

We can split the series as

$$f(x) = \underbrace{\sum_{i=0}^n \frac{f^{(i)}(c)}{i!} (x-c)^i}_{T_n(x)} + \underbrace{\sum_{i=n+1}^{\infty} \frac{f^{(i)}(c)}{i!} (x-c)^i}_{R_n(x)}$$

$$f(x) = T_n(x) + R_n(x) \quad f(x) - T_n(x) = R_n(x)$$

$$|f(x) - T_n(x)| = |R_n(x)|$$

Taylor's Inequality

If $|f^{(n+1)}(x)| \leq M$ for $|x-c| \leq d$, then the remainder $R_n(x)$ has the bound $|R_n(x)| \leq \frac{M}{(n+1)!} |x-c|^{n+1}$ where $|x-c| \leq d$.

$c=8$ center

Example: For $f(x) = \sqrt[3]{x}$ $T_2(x) = 2 + \frac{1}{12}(x-8) - \frac{1}{288}(x-8)^2$. How accurate is

$T_2(x)$ when $7 \leq x \leq 9$?

$$|x-8| \leq 1 = d \quad |R_2(x)| \leq \frac{M}{3!} |x-8|^3 \leq \frac{M}{6} \cdot 1^3 = \frac{M}{6}$$

Next: Find M so that $|f^{(3)}(x)| \leq M$ when $7 \leq x \leq 9$

$$f^3(x) = \left| \frac{10}{27} x^{-8/3} \right| = \frac{10}{27 x^{8/3}}$$

don't need $||$ for bound

$7 \rightarrow 9$ (all positive in this estn)

Since $x \geq 7 \rightarrow x^{8/3} \geq 7^{8/3} \rightarrow \frac{1}{x^{8/3}} \leq \frac{1}{7^{8/3}}$

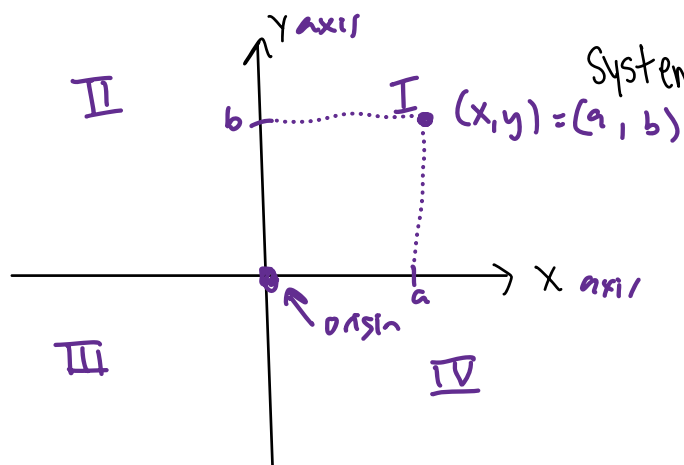
$$\rightarrow \frac{10}{27} \cdot \frac{1}{7^{8/3}} \leq \frac{10}{27 \cdot 7^{8/3}}$$

$$|R_2(x)| \leq \frac{M}{6} = \frac{.0021}{6} < \underline{0.0004 \text{ when } 7 \leq x \leq 9}$$

$$x^{8/3} = \overline{7}^{8/3} \rightarrow \overline{27x}^{8/3} \leq \frac{.0021}{\overline{27(7)}^{8/3}} = M$$

Three-Dimensional Coordinate Systems

Two-Dimensions



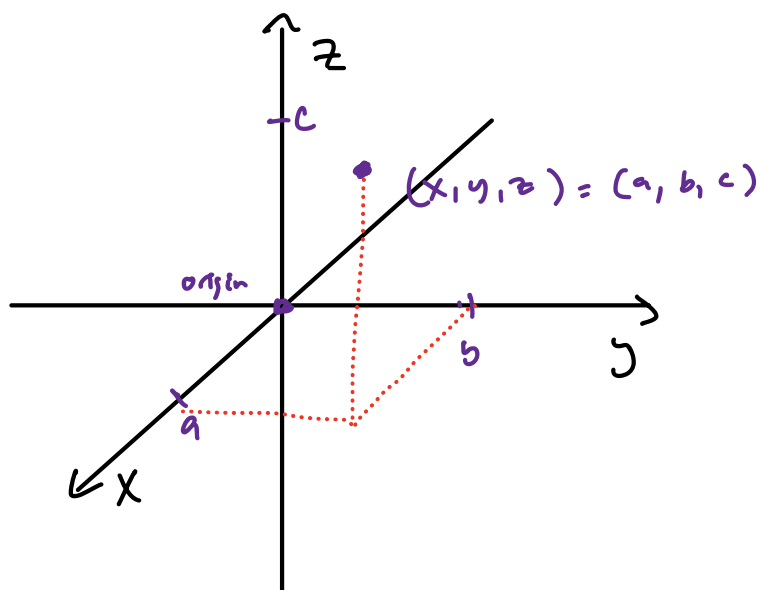
For the two-dimensional coordinate

System, we have an ordered pair (x, y)

- X-y plane or $\mathbb{R}^2$
with two coordinate axis:
X-axis & y-axis.

The first quadrant is where
 $x > 0$ and $y > 0$

Three-Dimensions:



$\mathbb{R}^3$

Activity: Plot the points $(1, 4, 5)$, $(0, -1, 5)$, and $(-2, 4, -1)$

Distance Formula

$$d = \sqrt{\underbrace{(x_2 - x_1)^2}_{\Delta x^2} + \underbrace{(y_2 - y_1)^2}_{\Delta y^2} + \underbrace{(z_2 - z_1)^2}_{\Delta z^2}}$$

Activity: Find the distance between $(1, 2, 3)$ and $(-2, 3, 0)$

Example: Are the points $A(0,1,0)$, $B(-1,2,3)$, and $C(5,1,6)$ collinear?

① Find $\overline{AB}$, $\overline{AC}$ and $\overline{BC}$

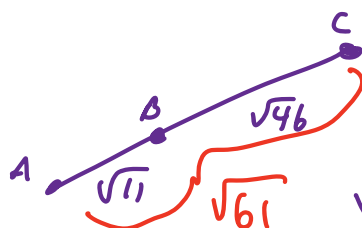
distance between
the two points

$$\overline{AB} = \sqrt{11}$$

$$\overline{AC} = \sqrt{61}$$

$$\overline{BC} = \sqrt{46}$$

If a, b, c were collinear, then:



$$\sqrt{11} + \sqrt{46} \neq \sqrt{61}$$

So A, B, C are NOT
collinear.

Equations in Three-Dimensions

Example: Graph the following in three-dimensions:

1.) $x = -2$

2.) $z = 0$

5.) $x + y = 4$

6.) $y^2 + z^2 = 4$